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Smart Gas Handling Strategies: A Case Study for Surplus Gas Management to Overcome the Challenge of Increased Gas Production

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Abstract

In mature oil reservoirs with rising Gas-Oil Ratios (GOR), surplus associated gas poses a growing challenge for gas handling facilities, particularly when gas volumes exceed the design capacities for reinjection and export. This paper presents a case study from a high- GOR onshore field where daily gas production has outpaced system constraints, creating a bottleneck that threatens oil production sustainability. Leveraging a structured approach combining historical production analysis, simulation using Aspen HYSYS, and technoeconomic screening, five strategic options were evaluated to mitigate the surplus gas issue. Key solutions included new compressor trains, compressor reconfiguration, rental compressor units, existing MP compressor enhancements, and high-pressure letdown strategy. A multi-criteria decision matrix evaluated each option's feasibility, cost, schedule, risk, and production impact. The letdown of HP discharge to the export network emerged as the most viable short-term solution due to its low capital cost, minimal modifications, and short implementation timeline. A phased implementation strategy was recommended, combining immediate mitigation with a scalable long-term new compression facility solution for the upcoming project.

The findings demonstrate how fit-for-purpose facility modifications can ensure production continuity, environmental compliance, and operational efficiency. The proposed methodology offers a practical and transferable model for brownfield facilities confronting similar gas-handling constraints in the context of energy transition and sustainable operations. This work also aligns with UAE's Net Zero 2050 commitments, emphasizing the role of efficient gas management in energy transition. In addition to production security, the mitigation of surplus gas avoids significant deferred oil revenues.

Introduction

In mature oil reservoirs, high Gas-Oil Ratios (GOR) present increasing operational challenges as fields age and associated gas production accelerates disproportionately to liquids. This phenomenon, particularly evident in gas cap expansion or depletion drive reservoirs, often leads to a surplus of produced gas that exceeds the designed capacity for both gas export and reinjection. In such scenarios, gas handling becomes a critical bottleneck. When the quantity of associated gas produced surpasses the maximum allowable

limits for export to pipeline networks and reinjection into the reservoir, operators face the pressing risk of production curtailment or regulatory noncompliance. This challenge is currently being confronted in one of Onshore fields, where surplus gas has begun to strain existing infrastructure, with no immediate capacity upgrades available.

The continuous increase in associated gas due to high GOR poses significant operational challenges across both subsurface and surface domains. One of the primary challenges is the rise in back pressure on the reservoir caused by bottlenecks in gas handling. This elevated pressure can lead to a measurable drop in oil production rates, especially in mature or pressure-depleted reservoirs, where even slight increases in flowing wellhead pressure can suppress production potential. Surface facilities, originally designed for lower GOR conditions, often lack the capacity or design flexibility to manage the excess gas volume, particularly when both gas export and reinjection systems are already operating at their mechanical limits. The situation is further exacerbated by the absence of design margin in existing process infrastructure to absorb sudden increases in gas flow due to process upsets, slugging, or reservoir behavior anomalies. As a result, frequent operational adjustments, venting, or flaring become unavoidable, increasing regulatory risk and carbon footprint. Additionally, the excessive gas load drives up compression power requirements, leading to higher utilities consumption and reduced system efficiency. These constraints not only threaten the sustainability of oil production but also compromise facility integrity and operational continuity, necessitating urgent and innovative gas management solutions.

Planning for Gas-Oil Ratio (GOR) is a challenging task due to several interrelated factors. Firstly, the unpredictable behavior of reservoirs makes it difficult to accurately forecast GOR evolution. This unpredictability stems from complex subsurface dynamics such as reservoir heterogeneity, gas cap movement, water influx, and well interference. These factors are particularly hard to model precisely, especially in older fields with limited data. Secondly, there is a delayed surface response to changes in GOR. Facility upgrades, such as compressors and pipelines, have long lead times of 12 to 36 months. By the time GOR increases are observed, it is often too late to implement large-scale infrastructure changes without impacting production. Thirdly, economic uncertainty adds another layer of complexity. Planning for GOR involves capital expenditure-heavy decisions, including new compressors, pipeline tie-ins, and rental units. These investments must be justified against uncertain production forecasts, fluctuating gas prices, and evolving ESG targets. Lastly, the integration of new systems into existing brownfield sites is inherently complex. Limited plot space, safety and shutdown risks, and utility system constraints all contribute to the challenge. Planning must balance short-term relief with long-term scalability, often under tight operational windows. These factors collectively make it difficult to predict and plan for GOR effectively.

Addressing this surplus gas is not just a matter of short-term operational optimization, it is a strategic imperative for ensuring sustainable oil production. Moreover, in the context of today's global energy transition and decarbonization agenda, intelligent gas utilization is central to both local and international mandates on emissions reduction, energy security, and resource efficiency. Converting surplus gas into value—rather than allowing it to become a liability—aligns directly with broader objectives of maximizing hydrocarbon recovery while minimizing environmental impact. It also supports corporate ESG goals and enhances the long-term economic viability of upstream operations, particularly in brownfields.

This study aims to identify and evaluate feasible, fast-track technical options to manage and monetize the surplus associated gas from the high GOR field without compromising ongoing production. The goal is to find a techno-economically viable solution that not only relieves pressure from existing gas systems but also integrates with sustainable energy practices. The focus is on delivering a practical strategy with minimal implementation lead time, scalable potential, and the flexibility to adapt to production variability. The paper presents the basis of study, screening methodology, and evaluation results for several options, ultimately recommending an integrated solution aligned with operational constraints, economic targets, and environmental standards. The study contextually supports global and local decarbonization goals by integrating surplus gas management into sustainable production frameworks.

Basis of Study

The basis of this study centers one of a mature onshore oil field currently experiencing elevated Gas-Oil Ratios (GOR). Historically, the field was developed with dual objectives: maximizing oil recovery and maintaining reservoir pressure through gas reinjection. Over time, due to gas cap expansion and production from high-GOR zones, gas production has risen sharply, while liquid output has plateaued or declined. Analysis of production trends revealed a consistent increase of 27% in daily gas production that constrained the gas handling facilities including the gas export compressor and the reinjection HP compressors under stable operating conditions. As a result, the field regularly experiences gas surplus events, especially during peak production or process upsets. The following figure represents the trend of oil production with associated GOR and water cut.

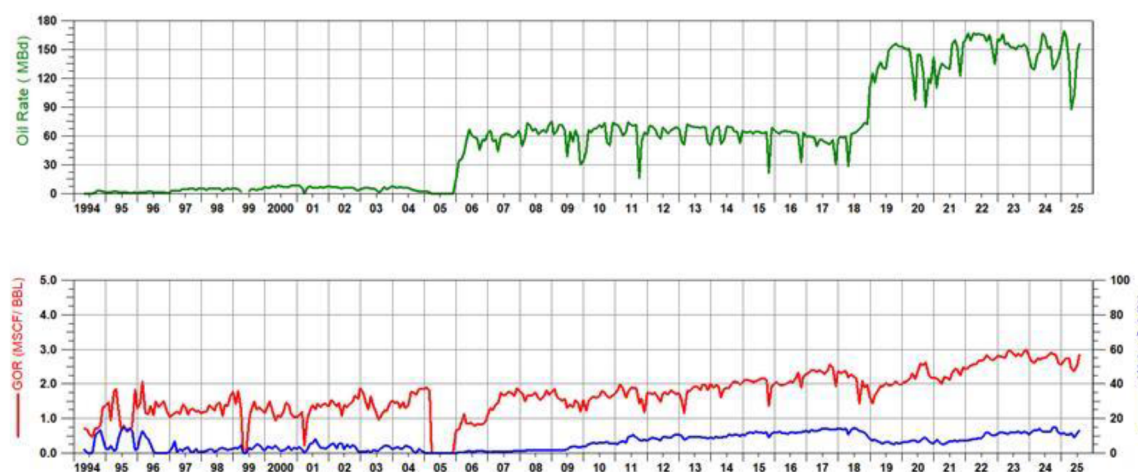


Figure 1—Oil Production, GOR and water cut

To quantify the challenge, historical production data was integrated into a detailed process simulation model using Aspen HYSYS. This model replicated the gas handling system, including low pressure (LP), medium pressure (MP) compression, gas dehydration units, high pressure (HP) and export gas (EG) compression train, and export lines. The simulation confirmed that both reinjection and export systems were operating near full capacity, with limited flexibility for surge conditions or further gas growth resulting in no design margin left the system vulnerable to shutdowns, increased back pressure on the reservoir, and ultimately, constrained oil production.

Reservoir surveillance data and predictive analytics for GOR forecasting were incorporated to strengthen integration of subsurface and surface studies.

Methodology

The study followed a structured approach, beginning with data consolidation and validation, followed by process modeling and scenario development. Key steps are outlined below:

Data Collection & Analysis: Historical well production, compression load trends, plant operating parameters, and gas reinjection/export profiles were collected. Production forecasts and GOR projections were also included to assess future gas handling requirements.

Process Simulation: A steady-state model of the surface facilities was developed and calibrated using recent field data. Simulation scenarios were run to test system responses to increasing gas loads, compressor outages, and gas export limitations.

Problem Definition: The simulation results confirmed that surplus gas volumes consistently exceed the system's combined handling capacity. Existing compressors are operating close to surge/stall conditions, with minimal margin to manage gas surges or process fluctuations.

Option Identification: Five potential mitigation strategies were identified:

- Installation of new export gas (EG) compressor trains.
- Reconfiguration of existing 2-stage HP compressors into parallel operation.
- Enhancement of MP/HP/EG compressor capacity via operating envelop optimization, debottlenecking or internals upgrade.
- Deployment of rental compression units as temporary relief solutions.
- Redirection of surplus gas to a nearby offtake point via tie-in to an adjacent facility's HP gas discharge pipeline.

Screening Criteria Development: To assess these options objectively, a multi-criteria decision matrix was established based on the following key drivers:

- **Technical Feasibility:** Compatibility with existing systems, reliability, and operational complexity.
- **Cost Effectiveness:** Estimated capital expenditure (CAPEX), operating expenditure (OPEX), and return on investment.
- **Implementation Time:** Procurement, fabrication, and installation lead times.
- **Operational Risk:** Potential for downtime, system stability, safety, and maintainability.
- **Production Gain or Impact:** Ability to sustain or enhance oil production by relieving gas constraints.

Preliminary Screening: Each option was ranked against the criteria using a weighted scoring system. Rental compression and HP discharge tie-in emerged as favorable due to fast implementation and minimal brownfield modification, although long-term sustainability was limited. New compressor trains offered robust capacity but came with high CAPEX and long lead times. MP/HP compressor enhancement and HP reconfiguration showed promising balance across all parameters but subject to compressor vendor's endorsement.

Detailed Evaluation: Shortlisted options underwent techno-economic evaluation, including detailed cost estimation, integration risk review, and production gain modeling. Sensitivity analyses were performed to account for variable gas growth rates and gas pricing assumptions.

Recommendation Development: A fit-for-purpose solution was recommended based on optimal alignment of technical, economic, and operational objectives. The selected path balances time of deployment, field production continuity, and minimal system disruption, ensuring both near-term mitigation and long-term gas management strategy. A multi-criteria decision matrix was utilized to ensure transparent and reproducible evaluation of all identified options. Risk management considerations, including flare minimization and HSE integrity, were explicitly integrated in the screening process.

Details of Options Evaluation

Each option was assessed based on its technical feasibility, ability to relieve the gas constraint, implementation timeline, integration complexity, and capital cost. The five options evaluated are described below in detail:

Option 1: Installation of a New Compressor Train

This option involves assessment for the installation of a dedicated new gas compressor train specifically designed to handle the incremental surplus gas and export it to the downstream export gas network. Technically, this provides a robust and scalable solution capable of matching the forecasted gas production profile. It allows full operational flexibility, independent of the reinjection system, and can be tailored precisely to handle the anticipated volumes.

However, this option presents significant implementation challenges. Firstly, the installation of a new compressor train requires additional plot space, which is not readily available on the existing onshore facility due to allocation to upcoming projects, necessitating potential civil and layout modifications. Integration with existing gas processing systems, particularly suction headers, discharge tie-ins, and control/shutdown logic modifications etc. requires complex engineering and potentially leading to partial shutdowns during tie-ins.

Furthermore, the new train would impose additional loads on the utility's infrastructure, including increased power demand and flare load capacity. These may trigger further system upgrades, adding to cost and time. The CAPEX for this option is high, and the project schedule is expected to be lengthy (typically 24-36 months) due to equipment procurement, fabrication, and commissioning timelines. While technically effective, the extended lead time and high cost reduce its attractiveness as a near-term mitigation measure. Strategically, this provides capacity for future tie-ins or regional third-party offtakes, supporting long-term gas infrastructure vision.

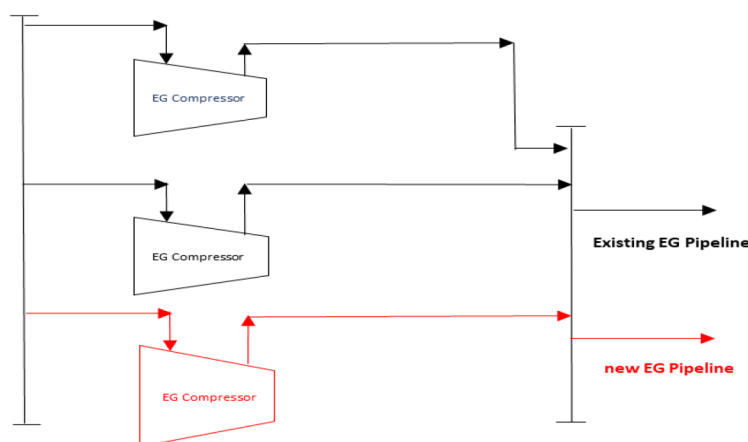


Figure 2—Option 1- New Compressor Train

Option 2: Enhancing the Capacity of Existing Medium Pressure (MP), High Pressure (HP) and Export Gas (EG) Compressor Trains

This option explores the debottlenecking or capacity enhancement of the existing medium pressure (MP), high pressure (HP) and export gas (EG) compressors through optimizing operating envelope, utilizing the design margins, modifications such as impeller replacement, casing re-rating, or motor upgrades. The intention is to increase throughput capacity to absorb surplus gas into the gas handling system. Leveraging spare capacity within existing assets can significantly reduce the need for future CAPEX, making it a critical factor in design and options evaluation. Monitoring spare operating and processing margins not only supports load optimization but also creates headroom for future expansion. Often underappreciated in energy efficiency assessments, CAPEX avoidance can yield far greater value than OPEX savings, as preserved capacity eliminates the requirement for new capital investments. Despite its apparent simplicity, this option has critical limitations. The maximum achievable capacity enhancement through such modifications is significantly lower than the forecasted surplus gas volumes. As a result, it does not offer a complete solution and would only partially alleviate the gas constraint. Moreover, increasing throughput in existing

machines leads to a corresponding rise in cooler duty, which the current system may not be able to handle effectively. There have also been historical issues with motor vibration and thermal imbalance during high-load operations, raising concerns over operational stability and long-term reliability.

Given these technical constraints and the inability to fully resolve the surplus gas issue, this option was screened out early in the evaluation process.

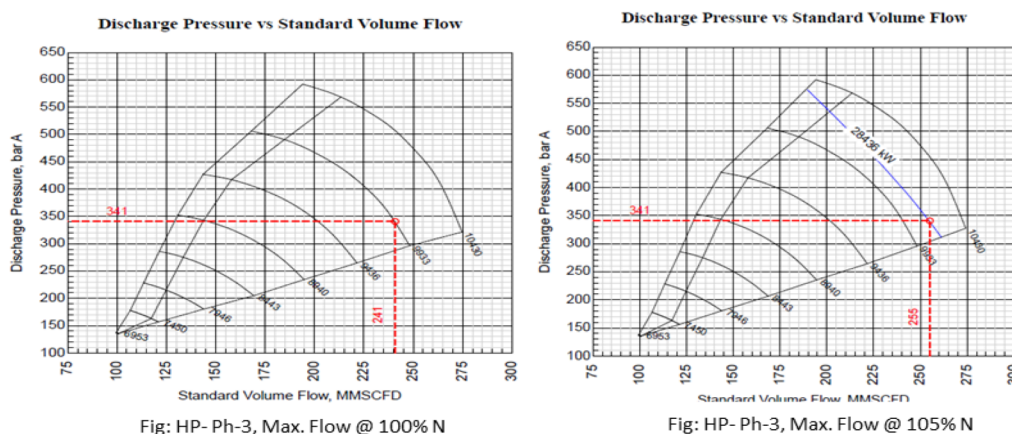


Figure 3—Option 2 - Compressor Capacity Enhancement

Option 3: Modification of 2-Stage HP Compressor Trains from Series to Parallel Configuration

In the existing setup, the high pressure (HP) compressors operate in a two-stage series configuration to compress gas for reinjection into the reservoir. However, with declining gas injection demand, a significant portion of compressor capacity is underutilized. This option proposes reconfiguring the compressors and associated equipment to operate from series to parallel configuration, allowing one train to be repurposed for gas export instead of injection. Compressor configuration is an important consideration in design and options evaluation. When assessing alternatives such as 1×100% versus 2×50%, both energy efficiency and CAPEX must be weighed against turndown flexibility under varying feed conditions. While split-capacity configurations can reduce recycle operation at low loads, their higher capital costs may outweigh the marginal energy savings, making a single-train configuration more practical in many cases.

This is a technically feasible and innovative reuse of existing assets. By modifying the train configuration, a portion of the HP compressor envelope can be matched to export pressure requirements, effectively using underutilized equipment to manage surplus gas. This configuration would improve operational flexibility and defer major new equipment CAPEX.

Nevertheless, the reconfiguration involves substantial brownfield modifications, including changes to suction and discharge piping, instrumentation and control logic, and shutdown systems. The most significant requirement is the construction of a 24 km export gas pipeline from the HP compressor discharge to the existing export pipeline network. This adds logistical and regulatory challenges, including land access, permitting, and tie-in risks. While the CAPEX is moderate compared to Option 1, the schedule remains lengthy due to brownfield complexity and pipeline construction, estimated at 18–24 months. Still, this option presents a strong medium-term solution with a good balance between cost and long-term utility.

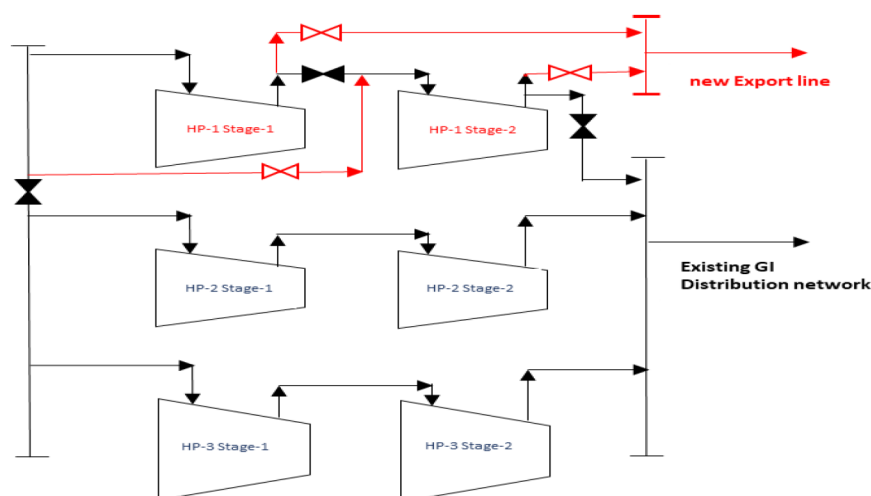


Figure 4—Option 4—Compressor Modifications from Series to parallel Configuration

Option 4: Deployment of Rental Compressor for Surplus Gas Export

To achieve rapid relief from gas constraints, this option considers deployment of temporary rental gas compressors to export surplus gas. The concept mirrors Option 1 but substitutes permanent equipment with rented units, typically containerized packages designed for quick mobilization and commissioning. This option offers an attractive implementation schedule typically within 12-16 months and avoids long procurement delays. It is particularly suitable for short-to-medium-term production relief while longer-term infrastructure solutions are developed.

However, the approach inherits several of the same technical and integration challenges as Option 1, including plot space, flare system validation, and tie-in works. Additional disadvantages stem from contractual and compliance considerations. Rental units must conform to the operating company's specifications, quality standards, and safety protocols, which may limit available suppliers or increase costs for customization. Long-term rental contracts may also prove more expensive than outright CAPEX investments in permanent equipment if used beyond 2–3 years. In summary, while rental compressors offer an expedited solution, their utility is best suited for interim mitigation rather than permanent resolution of the gas handling bottleneck but have the constraints of existing facilities integration, compliance to company standards and contractual requirements.

Option 5: Letdown of HP Discharge Gas to Align with Export Network Requirements

This option exploits the current operational observation that HP injection gas demand has reduced significantly due to changes in reservoir management strategy. Instead of reinjecting all gas into the reservoir, a portion of the HP discharge gas—already compressed above export line pressure—can be pressure-let down and routed to the export network.

This solution is technically simple and cost-effective, requiring minimal hardware modifications. It would involve installing a letdown station with control valves, overpressure protection, and tie-in piping to safely reduce HP discharge pressure to match export gas system specifications. Because the discharge gas is already dehydrated and processed, this route does not require additional conditioning or treatment. The CAPEX is low, and the implementation schedule is highly attractive, typically under 8-10 months. Moreover, this option leverages existing compressor infrastructure, requiring no major brownfield modifications or utility system upgrades. The primary risks are related to flow control, surge margin adjustments, and ensuring stability under dynamic operating conditions. Overall, this option emerges as a preferred short-term solution, offering a fast, low-cost pathway to relieve surplus gas while maintaining production and complying with export specifications. Indirect benefits include reduced flaring, improved energy efficiency, and alignment with carbon reduction strategies.

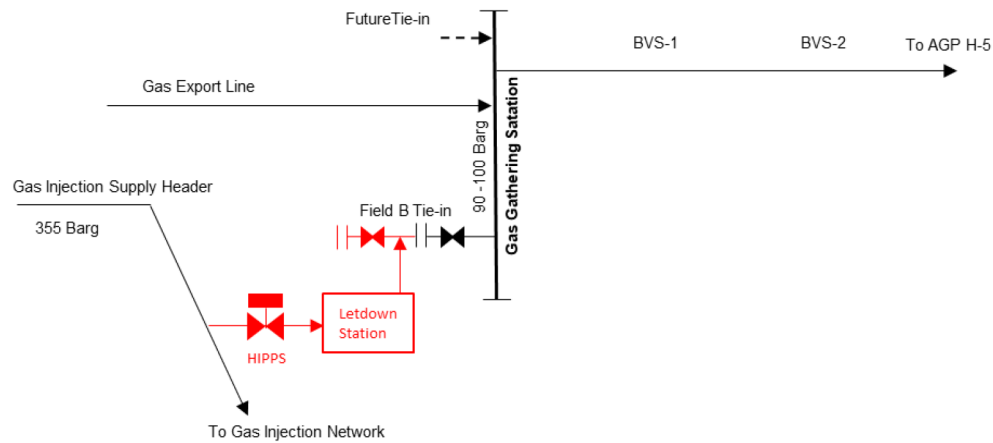


Figure 5—Option 5- High Pressure Gas Letdown

Screening Process

The screening of gas management options was conducted using a structured decision-making framework that incorporated both qualitative and quantitative parameters. Each of these options was ranked against key drivers (technical feasibility, cost, schedule, operational risk, and production impact) in the screening matrix, detailed in the next section. A **weighted scoring matrix** was developed, assigning relative importance to each criterion:

- Technical Feasibility (25%)
- Operational Risk (15%)
- Production Impact (15%)
- Implementation Time (30%)
- Cost Effectiveness (15%)

Each option was scored on a scale from 1 (least favorable) to 5 (most favorable) for each criterion. The weighted scores were then used to determine the overall ranking. The selection of the fit-for-purpose solution aims to strike the optimal balance between immediate relief, operational continuity, and future scalability. A radar chart was developed to visualize the comparative performance of each option across technical, economic, and risk dimensions.

Techno-Economic Results

A high-level techno-economic assessment was conducted for the four shortlisted options: New Compressor Train, HP Compressor Reconfiguration, Rental Compressor Deployment, and HP Discharge Letdown. Based on combined technical and economic analysis, Option 5 emerged as the most viable short-term solution, while Option 1 provides a sustainable long-term strategy to manage surplus gas within existing infrastructure.

Summary Table - Option Evaluation Comparison

Criteria	Option 1: New Compressor Train	Option 2: Enhance Existing Compressor	Option-3: HP Compressor Reconfiguration	Option 4: Rental Compressor	Option 5: Discharge Letdown
Technical Feasibility	High (5)	Low (2)	Moderate-High (4)	Moderate (3)	High (5)
Cost Effectiveness	Low (2)	Low (2)	Moderate (3)	Moderate (3)	High (5)
Implementation Time	Long (1)	Moderate (3)	Long (2)	Short (4)	Very Short (5)
Operational Risk	Moderate (3)	High (2)	Moderate (3)	Moderate-High (3)	Low (4)
Production Impact	High (5)	Low (1)	High (4)	Moderate (3)	High (4)
Total Weighted Score (/5.0)	3.10	1.90	3.55	3.25	4.60
Schedule (Est.)	24-30 months	12-18 months	18-24 months	6-8 months	<6 months
Status	Long-term Strategy	To be applied in parallel	Not Preferred	Short-term Relief	Recommended (Short-Term)

Conclusion

Based on comprehensive evaluation, the study recommends a phased implementation strategy combining a short-term relief measure with a long-term sustainable solution. The phased dual-track strategy combining Option 5 (short-term) and Option 1 (long-term) demonstrates innovative brownfield integration with maximum flexibility. Operational flexibility is enhanced by the ability to switch between letdown and reinjection depending on reservoir and offtake conditions.

Recommended Short-Term Solution: Option 5 - HP Discharge Letdown

The HP Compressor Discharge Letdown option emerged as the most practical and effective short-term solution. It offers the fastest implementation schedule (under ten months), requires minimal modifications to the existing infrastructure, and incurs low capital cost compared to other options. By utilizing the surplus HP compressor capacity, this strategy redirects gas from the injection line to the export line by reducing the discharge pressure through a letdown station. This option ensures immediate production continuity and eliminates the risk of curtailment due to gas handling limitations. It supports the company's sustainability goals by minimizing flaring and emissions while maintaining reservoir integrity and oil recovery efficiency.

Strategic Integration

A two-phase strategy ensures both immediate operational relief and long-term infrastructure resilience:

- Phase 1: Fast-track implementation of Option 5 (Letdown system) with Option-2 (Compressor capacity Enhancement).
- Phase 2: Parallel initiation of engineering and procurement activities for Option 1 (new compressor train), with target commissioning aligned to projected production growth.

This dual-track approach optimizes resources; mitigates gas handling risks, ensures compliance with environmental regulations, and enhances cost efficiency, aligning strongly with both field sustainability and the broader global energy transition.

Final Summary

High Gas-Oil Ratio (GOR) behavior in mature oil fields presents a significant operational and strategic challenge, particularly when the volume of associated gas exceeds the designed capacity of existing export

and reinjection systems. In this study, a structured approach was applied to evaluate multiple options for managing surplus gas at a brownfield facility constrained by gas handling infrastructure.

Through historical production analysis, process simulation, and facility modeling, it was confirmed that continued gas build-up would directly impact oil production due to rising reservoir backpressure and limitations in surface handling capacity. Five technically viable solutions were identified and screened based on feasibility, cost, timeline, operational risk, and production impact.

The **HP Discharge Letdown strategy (Option 5)** emerged as the most effective **short-term solution**, offering a rapid, low-cost implementation that requires minimal modification and delivers immediate production relief. For long-term sustainability, the new export gas compressor trains (Option 1) present a scalable and asset-efficient solution that aligns with future production trends and supports operational flexibility.

Together, these options form a phased gas management strategy that ensures continued production, maximizes infrastructure utilization, minimizes emissions, and aligns with broader sustainability and energy transition goals. The approach not only addresses the immediate surplus gas problem but also lays the foundation for future-proofing gas handling systems in a dynamic production environment. This methodology can serve as a benchmark for similar high-GOR field challenges in the region and globally. Societal benefits include enhanced energy security, reduced emissions, and improved cost efficiency in line with energy transition imperatives.

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